

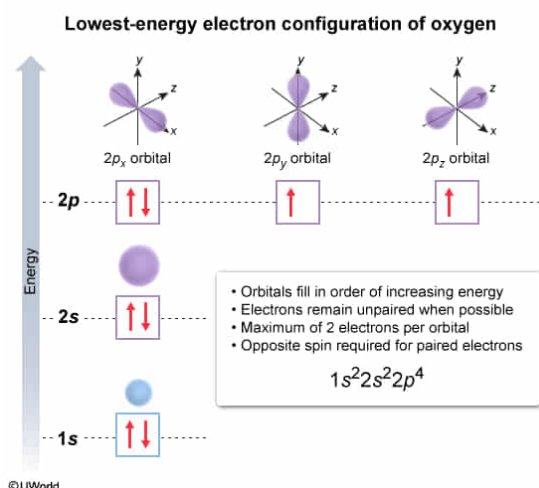
Which diagram below correctly represents the lowest-energy electron configuration of an oxygen atom?

- ☐ A. [8%]
- ☐ B. [4%]
- ☐ C. [1%]
- ☒ D. [84%]

Correct

84% answered correctly

Explanation:

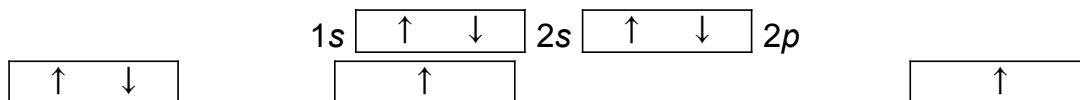


An **electron configuration** accounts for the placement of all electrons within the shells and subshells of an atom or ion. **Shells** are indicated using the principal quantum number n , which relates to the distance of an orbital from the nucleus. **Subshells** are labeled by type (**s**, **p**, **d**, or **f**), which relates to the shapes of the occupied orbitals present. **Electron configurations** are listed sequentially in **order** of increasing energy (**Aufbau principle**).

Moreover, the **Pauli exclusion principle** states that each orbital within a subshell can hold a maximum of two electrons, which must have opposite spins. In schematic diagrams, these

electrons are represented by arrows, and the spin of each electron is indicated by the orientation of the arrow (up or down). The orbitals containing the electrons are often represented by blanks or boxes. As the orbitals are filled, **Hund's rule** states that electrons remain unpaired until pairing is required to add another electron.

Following the Aufbau principle, the Pauli exclusion principle, and Hund's rule, the eight electrons in oxygen must have a configuration of $1s^2 2s^2 2p^4$ with a schematic representation of



(Choice A) This configuration violates Hund's rule. Two of the electrons in the $2p$ subshell should remain unpaired.

(Choice B) This configuration violates the Pauli exclusion principle. All paired electrons (including those in the $2p$ subshell) must have opposite spins.

(Choice C) This configuration violates the Aufbau principle. In the lowest-energy configuration, the lower energy orbitals (such as the $2s$ orbital) are filled before the orbitals of higher energy.

Educational objective:

An electron configuration accounts for all electrons held within the orbitals of an atom or ion, listed sequentially by shell and subshell in order of increasing energy following the Aufbau principle, Hund's rule, and the Pauli exclusion principle.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402309

System:
Atomic Theory and Chemical Composition